XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

Department of Electrical and Electronic Engineering

Xiamen University Malaysia

Quiz 1

COM105 Engineering Physics (I)

4 Questions

Total Marks (100)

26 April 2022 (Tuesday)

9:00 – 10:00 am (One hour)

INSTRUCTIONS TO CANDIDATES

1. Get your papers and pen ready, all corresponding working steps and answers **MUST be handwritten**. Make sure you have your data cable or have setup any cloud storage service app in your smartphone and PC (iCloud, Dropbox, Onedrive), which allows you to access your phone's photos or scanned images immediately from your PC.
2. This quiz consists of 4 main questions:

Q1:	1	sub-questions	Q2:	2	sub-questions
Q3:	2	sub-questions	Q4:	3	sub-questions

3. Please follow the requirements of each question, write your **ID, signature, and date** on the top right of **every** pages**.** Indicate the question number i.e., Q1a).
4. Scan your written answer papers with any phone scanner app of your choice, or take picture with the camera app and paste the images to an empty Word document. Make sure the images are clear and readable.
5. Compile all the images and save as **pdf** file. Name the pdf file as **COM105Quiz1_studentID.pdf**.
6. If the file size exceeded 20 MB limit, you may compile multiple **pdf** files according to the question number, as indicated in the file name i.e., **COM105Quiz1_studentID_Q1.pdf**, **COM105Quiz1_studentID_Q2&Q3.pdf**, etc. You are allowed to submit up to 4 **pdf** files in the Moodle Quiz submission tab.
7. You have extra 15 minutes to compile your answer sheet. Submission due at 10:15 am, after which will be considered late and late penalty will be applied.

Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you will be penalized accordingly.

Question 1

A drone flies with the following consecutive displacements: 3.50 m south, 8.20 m northeast, and 15.0 m west. Calculate the resultant displacement (magnitude and direction). Show your answers in three significant figures. **(25 marks)**

Question 2

A motorcyclist rides her motorcycle at 30.0 m/s and approaches a road sign at 100 m ahead. She then slows her motorcycle down with a constant acceleration of -3.50 m/s^2.

(a) Calculate the time taken for her to reach the road sign. **(20 marks)**

(b) Determine the velocity of the motorcycle when it reaches the road sign. **(5 marks)**

Question 3

In a snowball fight, Peter diverts Sean's attention by first throwing a high angle snowball at him, followed by a second snowball at a lower angle. Assume that both snowballs are thrown with a speed of 25.0 m/s, and the first one is thrown at an angle of 70.0° with respect to the horizontal.

(a) Find the angle of the second snowball to be thrown to reach Sean the same time as the first one. (Tips: $\sin 2\theta = 2\sin\theta\cos\theta$) **(20 marks)**

(b) Calculate the delay (in seconds) to throw the second snowball for it to reach Sean at the same time as the first. **(5 marks)**

Question 4

Three boys are applying forces on a box, given by F_1 = (-2.00i + 2.00j) N, F_2 = (5.00i - 3.00j) N, and F_3 = (-45.0i) N. When these forces are applied, the box accelerates with a magnitude of 3.75 m/s^2. Neglect friction.

(a) Determine the direction of the acceleration. **(10 marks)**

(b) Calculate the mass of the box. **(5 marks)**

(c) If the box is initially at rest, calculate its speed after 10.0 s. **(10 marks)**